Clustering Algorithms

Definition

Clustering is the process of dividing a dataset into groups (clusters) such that:

- Objects within the same cluster are similar.
- Objects in different clusters are dissimilar.
- No class labels are provided beforehand.

Need for Clustering

Clustering helps in:

- Customer segmentation
- Image segmentation
- Medical diagnosis
- Gene analysis
- Pattern recognition
- Market analysis

Characteristics of Clustering

1. Unsupervised learning technique.
2. Uses unlabeled data.
3. Groups similar objects together.
4. Helps discover hidden patterns in data.
5. Reduces complexity of large datasets.

Clustering Process

- Collect unlabeled data.
- Measure similarity between data objects.
- Apply clustering algorithm.
- Form clusters.
- Analyze and interpret clusters.

Types of Clustering Algorithms

1. K-Means Clustering

K-Means is the most popular partitioning clustering algorithm.

Steps

1. Select K cluster centers (centroids).
2. Assign each data point to the nearest centroid.
3. Recalculate centroids.
4. Repeat until centroids stop changing.

Advantages

- Simple and fast.
- Efficient for large datasets.
- Easy to implement.

Disadvantages

- Need to specify K beforehand.
- Sensitive to outliers.
- May converge to local optimum.

Example

Suppose student marks:

10, 12, 15, 80, 85, 90

For K=2:

- Cluster 1 → {10, 12, 15}
- Cluster 2 → {80, 85, 90}

2. Hierarchical Clustering

Creates a hierarchy of clusters represented as a tree structure called a **Dendrogram**.

Types

Agglomerative (Bottom-Up)

- Each object starts as its own cluster.
- Closest clusters are merged repeatedly.
- Continues until one cluster remains.

Divisive (Top-Down)

- Starts with one large cluster.
- Splits clusters recursively.

Advantages

- No need to specify number of clusters initially.
- Produces dendrogram for visualization.

Disadvantages

- Computationally expensive.
- Difficult for very large datasets.

Difference Between K-Means and Hierarchical Clustering

Feature	K-Means	Hierarchical
Type	Partitioning	Tree-based
Number of clusters	Required beforehand	Not required initially
Speed	Faster	Slower
Large datasets	Suitable	Less suitable
Output	Clusters	Dendrogram

Applications of Clustering

- Customer grouping in marketing
- Social network analysis
- Image processing
- Medical diagnosis
- Document classification
- Pattern recognition
- Recommendation systems

Advantages of Clustering

- Discovers hidden patterns.
- Works with unlabeled data.
- Useful for exploratory data analysis.
- Reduces data complexity.

Limitations of Clustering

- Results depend on similarity measures.
- Difficult to determine optimal number of clusters.
- Sensitive to noise and outliers.
- Different algorithms may produce different clusters.

Clustering Analysis is an unsupervised learning technique that groups similar objects into clusters. Objects within a cluster are highly similar and objects in different clusters are dissimilar. Common clustering algorithms are **K-Means** and **Hierarchical Clustering**. Clustering is widely used in market segmentation, image processing, pattern recognition, medical diagnosis, and recommendation systems. It helps discover hidden patterns in unlabeled data.

Clustering Analysis is a type of unsupervised learning used to divide data into groups called clusters. It aims to place similar objects in the same cluster and dissimilar objects in different clusters.

Types of Clustering Algorithms

1. K-Means Clustering
2. Hierarchical Clustering

K-Means Algorithm

- Choose K centroids.
- Assign points to nearest centroid.
- Recalculate centroids.
- Repeat until convergence.

Hierarchical Clustering

- Agglomerative (Bottom-Up)
- Divisive (Top-Down)
- Represented using a dendrogram.

Applications

- Customer segmentation
- Image segmentation
- Pattern recognition
- Medical diagnosis
- Recommendation systems

Advantages

- Finds hidden patterns.
- No labeled data required.
- Useful for exploratory analysis.

Limitations

- Sensitive to outliers.
- Choosing optimal clusters is difficult.
- Computationally expensive for large datasets.

Thus, clustering is an important unsupervised learning technique for discovering meaningful groups in data.

Differences Between Clustering and Classification

Clustering	**Classification**
Unsupervised learning	Supervised learning
No supervisor available	Supervisor available
Uses unlabelled data	Uses labelled data
No prior knowledge required	Domain knowledge required
Cluster results are dynamic	Labels usually remain fixed

Applications of Clustering

1. Grouping customers based on buying patterns.
2. Customer profiling based on lifestyle.
3. Information retrieval (finding documents from document collections).
4. Identifying groups of genes related to diseases.
5. Identifying organs with similar physiological functions.
6. Taxonomy of plants and animals in Biology.
7. Clustering based on purchasing behaviour and demographics.
8. Document indexing.
9. Data compression by grouping similar objects and identifying duplicates.

Challenges of Clustering Algorithms

1. High-Dimensional Data

- Modern datasets contain many features.
- Clustering becomes difficult when dimensions increase.

2. Large Volume of Data

- Internet generates billions of data records.
- Processing such huge datasets is challenging.

3. Scalability Problem

- Some algorithms work well on small datasets.
- The same algorithms may perform poorly on large datasets.

4. Different Units of Measurement

Example:

- Weight in kilograms (kg)
- Weight in pounds (lb)

Different units can affect clustering results.

5. Proximity Measure Design

- Choosing the correct similarity or distance measure is difficult.
- A poor distance measure leads to poor clusters.

Proximity Measures

What are Proximity Measures?

Clustering algorithms need a way to find how similar or different objects are so that they can be grouped into clusters.

- **Similarity** = How much two objects are alike.
- **Dissimilarity** = How much two objects are different.

These together are called **Proximity Measures**.

Usually, **distance measures** are used to find similarity between two objects i and j.

Distance Measures

Distance measures are also called **dissimilarity measures** because they indicate how different one object is from another.

Example:

- Smaller distance → More similar
- Larger distance → More different

Some measures such as **Cosine Similarity** directly indicate similarity between objects.

Distance between two objects i and j is denoted by:

$$D_{ij}$$

Properties of Distance Measures

1. Non-Negativity

$$D_{ij} \geq 0$$

- Distance is always positive or zero.
- Distance can never be negative.

2. Identity Property

$$D_{ii} = 0$$

3. Symmetry Property

$$D_{ij} = D_{ji}$$

Distance from object i to j is the same as distance from j to i

Example:

Distance(A,B) = Distance(B,A)

4. Triangle Inequality

$$D_{ij} \leq D_{ik} + D_{kj}$$

- Direct distance between two objects should be less than or equal to the distance through another object.

Example:

$A \rightarrow B \leq A \rightarrow C + C \rightarrow B$

Metric Distance

If all four properties are satisfied, then the distance measure is called a **Metric**.

Distance Measures Based on Data Types

Distance measures depend on the type of data used.

Data types are mainly divided into:

1. Categorical Variables

Values represent categories.

Example:

Gender = {Male, Female}

These are called **Nominal Variables**.

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2. Ordinal Variables

These are similar to nominal variables but have an inherent order.

Example:

Temperature = {Low, Medium, High}

Order exists:

$$Low < Medium < High$$

3. Quantitative Variables

These contain numerical values.

Examples:

- Real numbers
- Integer numbers

Example:

- Age = 20
- Salary = 30000

4. Binary Data

Attributes can take only two values.

Example:

- Yes / No
- True / False
- 1 / 0

Objects having binary attributes are called Binary Objects.

Quantitative Variables

Quantitative variables are numerical variables. Distance measures are commonly used for these variables.

1. Euclidean Distance

- Euclidean distance is the most common distance measure.
- It is also called **L_2 norm**.
- It is the square root of the sum of squared differences between coordinates of two objects.

Formula

$$D(x_i, x_j) = \sqrt{\sum_{k=1}^{n}(x_{ik} - x_{jk})^2}$$

Advantages

- Distance does not change when new objects are added.

Disadvantages

- If units change, the distance changes significantly.
- Computational cost is high because of square and square root operations.

2. City Block Distance (Manhattan Distance)

Also called:

- Manhattan Distance
- Taxicab Distance

- Boxcar Distance
- Absolute Value Distance
- L_1 norm

Formula

$$D(x_i, x_j) = \sum_{k=1}^{n} |x_{ik} - x_{jk}|$$

- Distance is calculated by adding the absolute differences of coordinates.

3. Chebyshev Distance

Also called:

- Maximum Value Distance
- Supremum Distance
- $L\infty$ norm

Formula

$$D(x_i, x_j) = \max_k |x_{ik} - x_{jk}|$$

Take the maximum difference among all coordinates.

Partitional Clustering Algorithm (K-Means)

Introduction

- **K-means** is a simple and iterative partitional clustering algorithm.
- Here, **k** represents the number of clusters specified by the user.
- The algorithm assumes that clusters do **not overlap**.
- A data point can belong to **only one cluster**.
- It can detect clusters having **circular or spherical shapes**.

Initialization

- The algorithm starts by selecting **k data points** randomly as initial cluster centers (centroids).
- Prior knowledge is usually absent.
- Therefore, choosing good initial centroids is important.
- Data points should be **normalized** for better performance.

Working of K-Means

The main objective is to assign each data point to the nearest cluster centroid.

The process continues until no further changes occur in cluster assignments.

Steps of K-Means Algorithm

Step 1

Determine the number of clusters (**k**).

Step 2

Choose **k instances randomly** as initial cluster centers.

Step 3

Compute the centroid and assign each sample to the nearest cluster using Euclidean distance.

Step 4

Re-compute the centroid using newly assigned samples.

Step 5

Repeat Steps 3 and 4 until no changes occur in cluster assignments.

Sum of Squared Error (SSE)

K-means attempts to minimize the **Sum of Squared Error (SSE)**.

$$SSE = \sum_{i=1}^{k} dist(c_i, x)^2$$

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Where:

- c_i = centroid of cluster i
- x = data point
- $dist$ = Euclidean distance

Aim

Minimize SSE.

Advantages

1. Simple

2. Easy to implement

Disadvantages

1. Sensitive to initialization; different starting points may produce different clusters.
2. For large datasets, execution time increases.

How to Choose the Value of k?

- There is no fixed rule for selecting **k**.
- Run K-means for different values of **k**.
- Compute within-group variance (sum of squares).
- Plot the results as a graph called the **Elbow Curve**.
- The point where the graph becomes almost flat indicates the optimal value of **k**.

Complexity

The complexity depends on:

- n = number of samples
- k = number of clusters
- I = number of iterations
- d = number of attributes

$$\Theta(nkId)$$

The complexity of K-means algorithm is:

$$O(n^2)$$

Example 13.5: Consider the following set of data given in Table 13.9. Cluster it using *k*-means algorithm with the initial value of objects 2 and 5 with the coordinate values (4, 6) and (12, 4) as initial seeds.

Table 13.9: Sample Data

Objects	X-coordinate	Y-coordinate
1	2	4
2	4	6
3	6	8
4	10	4
5	12	4

Initial Seeds

Object 2 = (4,6)

Object 5 = (12,4)

Thus:

Cluster 1 Centroid (4,6)

Cluster 2 Centroid (12,4)

Iteration 1

Object 1 = (2,4)

Distance to Centroid 1:

$$\sqrt{(2-4)^2+(4-6)^2}$$
$$=\sqrt{8}$$

Distance to Centroid 2:

$$\sqrt{(2-12)^2+(4-4)^2}$$
$$=\sqrt{100}$$
$$=10$$

Therefore, Object 1 belongs to Cluster 1.

Object 3 = (6,8)

Distance to Centroid 1:

$$\sqrt{(6-4)^2+(8-6)^2}$$
$$=\sqrt{8}$$

Distance to Centroid 2:

$$\sqrt{(6-12)^2+(8-4)^2}$$
$$=\sqrt{52}$$

Object 3 belongs to Cluster 1.

Object 4 = (10,4)

Distance to Centroid 1:

$$\sqrt{(10-4)^2+(4-6)^2}$$
$$=\sqrt{40}$$

Distance to Centroid 2:

$$\sqrt{(10-12)^2+(4-4)^2}$$
$$=\sqrt{4}$$
$$=2$$

Object 4 belongs to Cluster 2.

Object 5 = (12,4)

Already belongs to Cluster 2.

Cluster Table After Iteration 1

Cluster 1

- (4,6)
- (2,4)
- (6,8)

Cluster 2

- (10,4)
- (12,4)

New Centroids

Cluster 1

$$\left(\frac{4+2+6}{3}, \frac{6+4+8}{3}\right)$$

$$= (4, 6)$$

Cluster 2

$$\left(\frac{10+12}{2}, \frac{4+4}{2}\right)$$

$$= (11, 4)$$

Iteration 2

Object 1 (2,4)

Distance to Cluster 1:

$$\sqrt{8}$$

Distance to Cluster 2:

$$\sqrt{81}$$
$$= 9$$

Remains in Cluster 1.

Object 3 (6,8)

Distance to Cluster 1:

$$\sqrt{8}$$

Distance to Cluster 2:

$$\sqrt{41}$$

Remains in Cluster 1.

Object 4 (10,4)

Distance to Cluster 1:

$$\sqrt{40}$$

Distance to Cluster 2:

$$\sqrt{1}$$

$$= 1$$

Remains in Cluster 2.

Object 5 (12,4)

Distance to Cluster 1:

$$\sqrt{68}$$

Distance to Cluster 2:

$$\sqrt{1}$$

$$= 1$$

Remains in Cluster 2.

Final Cluster Table

Cluster 1

- (4,6)
- (2,4)
- (6,8)

Centroid = (4,6)

Cluster 2

- (10,4)
- (12,4)

Centroid = (11,4)

Result

No change occurs in the second iteration.

Therefore, the **K-means algorithm terminates**.

Final Clusters

Cluster 1: {(2,4), (4,6), (6,8)}

Cluster 2: {(10,4), (12,4)}

Example 8.4

Assess a Student's performance. Using the Naive Bayes algorithm for continuous attributes.

Predict whether a student get a job offer or not in the final year using the training dataset T consisting of 10 data instances with attributes such as "CGPA" and "Interactiveness" shown in Table 8.13. The target variable is job offer, which is classified as yes or No. for a Student.

Training Dataset

Sl No	CGPA	Interactiveness	Job offer
1	9.5	yes	yes
2	8.2	No	yes
3	9.3	No	NO
4	7.6	No	No
5	8.4	yes	yes
6	9.1	yes	yes
7	7.5	yes	NO
8	9.6	No	yes
9	8.6	yes	yes
10	8.3	yes	yes

Test Data

Predict the job offer for a student having: CGPA = 8.5 and Interactiveness = yes using Gaussian Naive Bayes.

Step 1: Prior Probability

From the table:

Job offer	count
Yes	7
NO	3

Therefore:

$$P(\text{JobOffer} = \text{Yes}) = \frac{7}{10}$$

$$P(\text{Job offer} = \text{No}) = \frac{3}{10}$$

Step 2: Interactiveness probabilities.

For Job offer = yes

Out of 7 yes records:

* Interactive = Yes → 5

Therefore:

$$P(\text{Interactive} = \text{yes} \mid \text{Job Offer} = \text{yes}) = \frac{5}{7}$$

for Job offer = No

Out of 3 No records

* Interactive = yes → 1

Therefore:

$$P(\text{Interactive} = \text{yes} \mid \text{Job offer} = \text{No}) = \frac{1}{3}$$

Step 3: Continuous Attributes (CGPA)

For yes class,

1. Find μ_{yes} (Mean for Job offer = yes)

Take only the CGPA values where Job offer = yes:

CGPA

9.5

8.2

8.4

9.1

9.6

8.6

8.3

formula

$$\mu = \frac{\Sigma x}{N}$$

Adding all values: $9.5 + 8.2 + 8.4 + 9.1 + 9.6 + 8.6 + 8.3$

$= 61.7$

No of values $= 7$

$$\mu_{yes} = \frac{61.7}{7} = 8.814286$$

Q Find σ_{yes} (Standard Deviation)

Formula

$$\sigma = \sqrt{\frac{\Sigma (x-\mu)^2}{N}}$$

Using $\mu = 8.8143$

x	$x-\mu$	$(x-\mu)^2$
9.5	0.6857	0.4702
8.2	−0.6143	0.3774
8.4	−0.4143	0.1716
9.1	0.2857	0.0816
9.6	0.7857	0.6173
8.6	−0.2143	0.0459
8.3	−0.5143	0.2645

Add the last column: 2.0285

Variance: $\frac{2.0285}{7} = 0.2898$

Standard deviation:

$$\sigma_{yes} = \sqrt{0.2898}$$

$$= 0.5383$$

3. Find μ_{NO}.

Take only Job Offer = No:

CGPA

9.3

7.6

7.5

Add: $9.3 + 7.6 + 7.5 = 24.4$

Divide by 3:

$$\mu_{NO} = \frac{24.4}{3} = 8.1333$$

4. Find σ_{NO}.

Using $\mu = 8.1333$

x	$x - \mu$	$(x-\mu)^2$
9.3	1.1667	1.3611
7.6	-0.5333	0.2844
7.5	-0.6333	0.4011

Add Squares $1.3611 + 0.2844 + 0.4011 = 2.0466$

$$\text{variance} = \frac{2.0466}{3} = 0.6822$$

Standard deviation:

$$\sigma_{NO} = \sqrt{0.6822}$$

$$= 0.826$$

Step 4: Calculate Gaussian probability

Test CGPA:

$x = 8.5$

Use Gaussian formula:

$$P(x = x/C) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

For yes

Substitute $x = 8.5$

$\mu = 8.8714$

$\sigma = 0.38146$

$$P(CGPA = 8.5 \mid yes) = 0.597$$

For No.

Substitute $x = 8.5$

$\mu = 8.1333$

$\sigma = 1.0116$

$$P(CGPA = 8.5 \mid No) = 0.369$$

Step 5: Apply Naive Bayes

For YES

$$P(yes) \times P(CGPA/yes) \times P(Interactive = yes)$$

$$= \frac{7}{10} \times 0.594 \times \frac{5}{7}$$

$$= 0.297$$

$$=$$

Step 6: for NO

$$P(No) \times P(CGPA|No) \times P(Interactive=Yes|No)$$

$$= \frac{3}{10} \times 0.369 \times \frac{1}{3}$$

$$= 0.0369$$

Step 7: Compare

class	probability
Yes	0.297
No	0.0369

Since $0.297 > 0.0369$

Final Answer

Job Offer = Yes